

INTERNSHIP REPORT PRESENTATION

Python Developer Internship

Codec Technologies

1-Month Internship • Python Development

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Candidate Profile

**[YOUR
NAME]**

[B.Tech / BCA / MCA]
Computer Science



Roll Number

25SCS1003005701



Institute

IILM University Greater Noida



Programme

B.Tech CSE



Internship Domain

Python Development



Duration

1 Month

Codec Technologies — Python Developer Internship

The one-month internship is structured around practical Python development, moving from fundamentals and data structures to advanced Python, web development, data handling, APIs, debugging and a final project.

01

Python Foundations

Syntax, variables, data types, control flow, functions, exceptions and core data structures.

02

Advanced Development

OOP, modules, packages, file handling, decorators, generators and Flask web development.

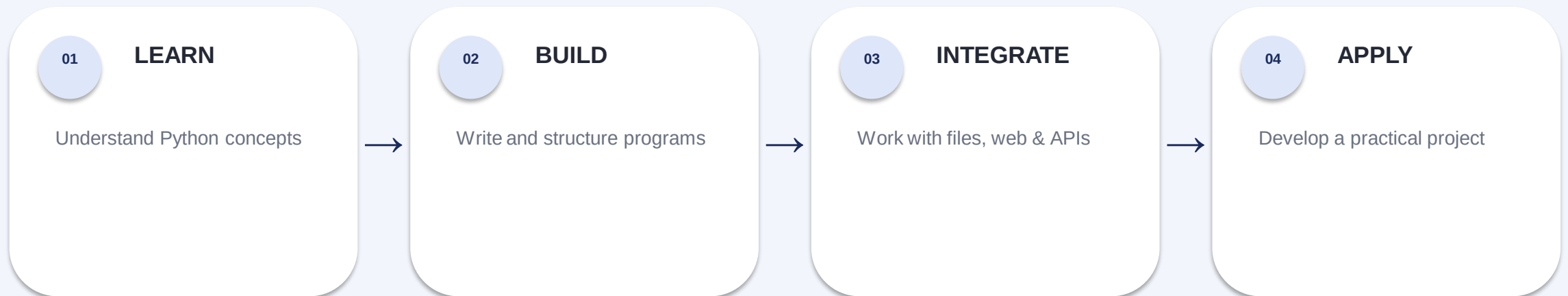
03

Data, APIs & Project

Pandas, data cleaning, visualization, REST APIs, requests, debugging and a final project.

Bridging Theory and Practical Python Development

Academic learning often introduces programming concepts separately. The internship provided a structured path to connect those concepts into practical development workflows.



What the Internship Set Out to Achieve

PY

Python Fundamentals

Build a strong foundation in Python syntax, control flow, functions and exceptions.

DS

Data Structures

Work confidently with lists, tuples, sets, dictionaries and comprehensions.

OP

Advanced Python

Understand OOP, modules, packages, files and selected advanced concepts.

FL

Web Development

Learn Flask basics, routes, requests and templates.

API

Data & APIs

Use Pandas and integrate REST APIs with HTTP requests.

PR

Practical Project

Apply the learned concepts to a simple internship project.

The Toolkit Behind the Internship



Python

Core programming language



VS Code / IDE

Development and execution environment



Flask

Web application framework



Pandas

Data analysis and manipulation



Requests

HTTP and API integration



CSV / JSON

Structured data and file handling



Git / Version Control

Use if applicable to your work



Project Tools

Add any tools actually used

From Input to Working Application

Use this as the high-level workflow for the internship project. Replace labels with your exact project flow if needed.



The 4-Week Learning Journey



API Performance Monitor

Project description: A Python-based web application that monitors API performance by measuring response time, tracking HTTP status codes, recording successful and failed requests, and displaying performance metrics through a Flask dashboard.

01

Objective

To develop a monitoring system that helps analyze API reliability and response performance.

02

Technologies

Python • Flask • Requests • SQLite • HTML/CSS

03

Key Features

- API endpoint testing
- Response-time measurement
- HTTP status monitoring
- Success/failure tracking
- SQLite logging
- Performance metrics dashboard

API Performance Monitor

Flask + Requests + SQLite

Codec Technologies Internship Project

Test an API Endpoint

Check APIClear Logs

Enter an endpoint and click "Check API".

REQUESTS

5

SUCCESSFUL

4

FAILED

1

AVG RESPONSE

1510.9 ms

SUCCESS RATE

80.0%

Recent API Checks

URL	Status	Response	Result	Time
https://httpbin.org/delay/2	200	3267.13 ms	SUCCESS	2026-09-10 20:19:08
https://httpbin.org/status/200	200	1259.55 ms	SUCCESS	2026-09-10 20:18:56
https://httpbin.org/status/404	404	1306.62 ms	FAILED	2026-09-10 20:18:36
https://jsonplaceholder.typicode.com/todos/1	200	436.63 ms	SUCCESS	2026-09-10 20:18:26
https://httpbin.org/get	200	1284.55 ms	SUCCESS	2026-09-10 20:18:15

Response-Time Summary

Fastest

436.63 ms

Slowest

3267.13 ms

80.0% of recorded requests were successful.

Skills & Capabilities Gained

- Stronger understanding of Python programming fundamentals
- Practical use of data structures, functions and OOP
- Experience with file handling and structured data
- Exposure to Flask and basic web application development
- Understanding of Pandas for data handling
- Experience consuming and processing REST API data
- Improved debugging and problem-solving ability
- Ability to apply concepts in a practical project

COMPLETION

Successfully Completed

Python Developer Internship

Organization

Codec Technologies

Duration

1 Month

Certificate / ID

CORPORATE6759d549ce59e1733940553



Thank You